

# Guarded Current-Evidence Fusion for Cooperative Multitarget Tracking

**Abstract**—Cooperative multi-object tracking must distinguish new observations from information inherited through earlier communication. Posterior age weighting acts on both components together, limiting control over current sensing evidence. We propose guarded current-evidence fusion (GCE), which augments conservative geometric pooling with bounded Bernoulli posterior-to-prediction ratios. An admission rule combines current sensing opportunity, association-weighted detection-score and miss support, and Gaussian curvature to select local information increments. A closed-form Gaussian implementation jointly updates spatial density and target existence through a shared normalizer, allowing admitted evidence to influence both localization and cardinality. Across 43 V2V4Real segments evaluated with a common local tracker, GCE reduces mean OSPA by 5.8% and 4.4% relative to qualified geometric pooling (No-age KLA) under reliable and intermittent links, respectively. An exact zero-omission code preserves GCE trajectories while reducing fragmented traffic by 11.3% and 10.8% relative to full GCE packets on the original 25-segment comparison corpus.

## I. INTRODUCTION

Cooperative robots observe different targets, occlusions, and missed detections. Their exchanged tracks also contain information inherited from earlier communication. A recently received posterior may therefore carry old evidence, while a low existence probability may convey a useful current miss. Effective fusion must distinguish inherited beliefs from new observations without transmitting the entire sensing history.

Labeled multi-Bernoulli (LMB) filtering represents each track by a label, existence probability, and conditional spatial density [1], [2]. Geometric pooling, or Kullback–Leibler averaging (KLA), combines local posteriors when their cross-correlations are unavailable [3]. In the ideal shared-prior case, normalized geometric weights retain one copy of the prior but also temper each new likelihood. Changing posterior weights according to observation age does not separately control these contributions. A further coupling arises within each Bernoulli: the integral of its spatial product affects the probability that the target exists. An existence correction alone leaves this part of the current evidence unused. Figure 1 illustrates the distinction: a posterior weight acts on inherited and current information together, whereas a current ratio can change their relative contributions.

We propose *guarded current-evidence fusion* (GCE). Each robot encodes the ratio between its local update and prediction. Fusion augments a conservative posterior pool with admitted fractions of these ratios, then jointly normalizes existence and spatial density. Current sensing opportunity, association-weighted score support, and Gaussian curvature determine which increments are admitted. Figure 2 shows

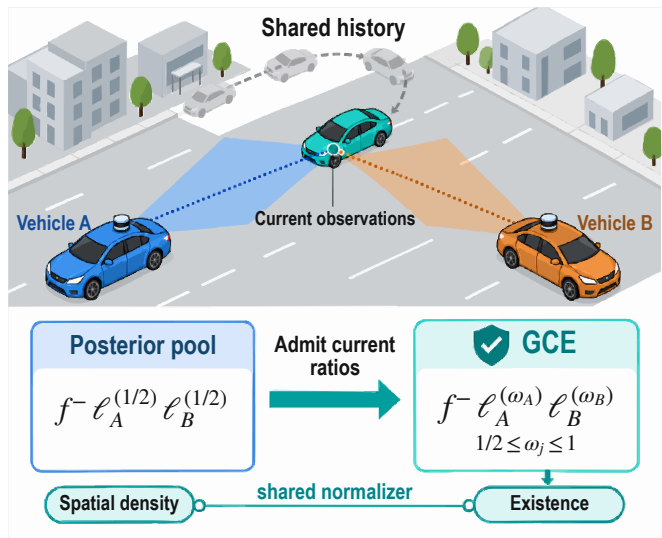


Fig. 1. Cooperative tracking with shared history (illustration). In the idealized shared-prior case, equal-weight geometric pooling retains the prior but tempers each current likelihood. GCE admits bounded fractions of current posterior-to-prediction ratios, increasing the likelihood exponents  $\omega_j$  while retaining one prior. Joint normalization carries this correction into both spatial density and existence.

how the same current evidence reaches both branches of the fused belief.

Building on prior–likelihood separation [4], [5], GCE makes selective admission of approximate local increments the central fusion decision when sensing histories and associations differ. Bernoulli normalization then carries each admitted increment into both parts of the track belief. The paper contributes:

- A bounded admission rule combining current opportunity, association-weighted score and miss support, and a source-level curvature check for approximate Gaussian Bernoulli ratios.
- A closed-form implementation that couples admitted spatial and existence corrections, with an exact omission code for zero vectors.
- A shared-tracker evaluation with 5.8%/4.4% lower mean OSPA than No-age KLA across 43 V2V4Real segments, supported by matched recursive controls and explicit communication accounting.

## II. RELATED WORK

### A. Cooperative robotic tracking

Distributed probability hypothesis density (PHD) filters support robot search [6], decentralized action selection [7],

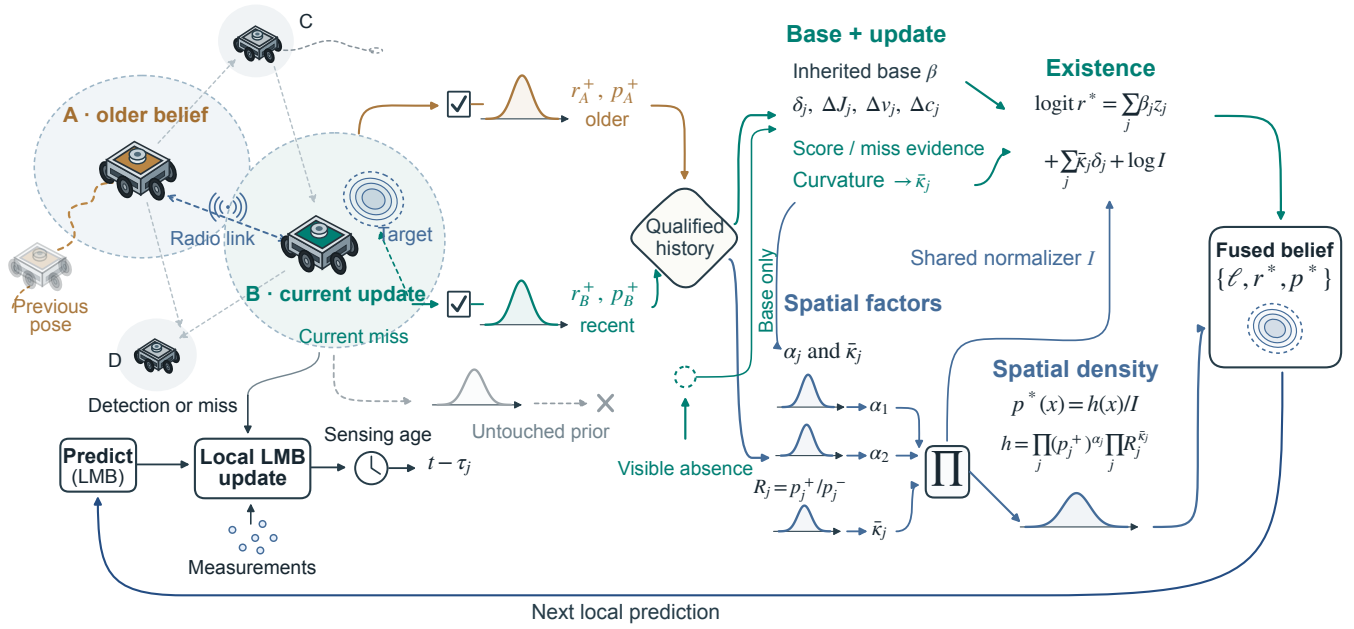


Fig. 2. Guarded current-evidence fusion. Robots exchange inherited beliefs and current Bernoulli ratios. Qualified support forms the base; sensing, score, and curvature determine the admitted correction. The same ratio updates spatial density and existence through the shared integral  $I$ . The fused belief becomes the next local prior. Robot poses and density shapes illustrate the mechanism.

and sensing reconfiguration [8]; LMB filters also support distributed mapping [9]. V2V4Real provides real cooperative driving observations [10], and DMSTrack learns uncertainty in a differentiable multisensor tracker [11].

Where2comm selects spatial features by confidence [12]; CoopTrack exchanges instance features and learns cross-agent association [13]. GCE operates on probabilistic track summaries using DMSTrack’s per-vehicle detections.

### B. Finite-set fusion and sensing support

Covariance intersection addresses fusion with unknown cross-correlations [14], and KLA formulates geometric pooling over probability densities [3]. Generalized covariance intersection (GCI) extends geometric pooling to PHD filters [15]; consensus labeled random finite set (RFS) filters preserve track identities [16].

Direct GCI fusion can degrade under inconsistent labels [17]; linear assignment can align local identities before labelwise fusion [18]. Different fields of view require explicit treatment of absent labels and exclusive information [19]–[21]. Minimum-information-loss (MIL) fusion uses arithmetic pooling of labeled finite-set densities [22]; track consensus matches and combines short trajectories [23]. These methods motivate our use of aligned labels and explicit sensing support.

Cardinality and localization also play distinct roles in finite-set fusion: pointwise density consistency alone does not ensure a useful cardinality distribution [24]. Heterogeneous fusion assigns confidence at the component level [25]. GCE instead couples the two components through the normalizer of an admitted Bernoulli update.

### C. Prior and current likelihood information

Shared-prior Bayesian fusion corrects for common information in local posteriors [4]. Likelihood consensus aggregates exponential-family likelihood representations [5]. GCE transmits Gaussian natural-parameter increments and admits them fractionally. Local LMB association and moment projection can make these ratios differ from exact sensor likelihoods. The admission rule is a practical restriction on such approximations; it does not estimate unknown cross-source correlation or certify covariance consistency.

## III. GUARDED CURRENT-EVIDENCE FUSION

### A. Local prediction, update, and information support

Each robot predicts its LMB, updates it with the current local measurement set, and exchanges one posterior packet with available neighbors. Fusion output becomes the next local prior. A label is aligned across sources before fusion; births remain local and measurement driven. For a represented label, write the local prediction and update as  $f_j^- = (r_j^-, p_j^-)$  and  $f_j^+ = (r_j^+, p_j^+)$ . The Bernoulli density has empty-set mass  $1-r$  and singleton density  $rp(x)$ . The implementation moment-projects  $p_j^-$  and  $p_j^+$  to four-dimensional Gaussians over planar position and velocity.

A direct sensing opportunity occurs when the modeled detection probability at a predicted track is positive. Both a detection and a missed detection advance the local direct-opportunity timestamp  $\tau_j$ ; receiving another robot’s packet does not advance it. A separate history flag records whether a represented belief descends from an informative local update. When informed inputs exist, untouched priors are excluded. An absent label contributes an existence censor only if its source could observe the represented target. Otherwise

that source abstains. Censors carry no spatial density. This support handling is shared with the qualified no-age and recency controls.

Let  $\mathcal{P}$  be eligible represented sources and  $\mathcal{E}$  eligible existence sources, including qualified censors. Normalize ordinary delivered weights  $w_j$  separately on the two supports:

$$\alpha_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad b_j = \frac{w_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k}. \quad (1)$$

All sums below use the corresponding support. A label without spatial support is omitted. For a represented source, define

$$a_j = \rho + (1 - \rho)e^{-(t-\tau_j)\Delta t/\lambda}, \quad q_j = \frac{b_j a_j}{\sum_k b_k a_k}, \quad (2)$$

with  $a_j = \rho$  before any local opportunity and  $a_j = 1$  for a censor.

### B. A conservative inherited base

Set  $z_j = \text{logit}(r_j^+)$  and choose the existence weights

$$\beta = \begin{cases} q, & \sum_j (q_j - b_j) z_j < 0, \\ b, & \text{otherwise.} \end{cases} \quad (3)$$

Thus history can lower the base existence log-odds but cannot raise them. The spatial base retains the ordinary weights  $\alpha$ . An unnormalized base has masses

$$A_0 = \prod_j (1 - r_j^+)^{\beta_j}, \quad (4)$$

$$A_1(x) = \prod_j (r_j^+)^{\beta_j} \prod_j p_j^+(x)^{\alpha_j}. \quad (5)$$

These are separate existence and spatial weights; when the weights and supports coincide, the base reduces to ordinary Bernoulli KLA. The no-history ablation sets  $\beta = b$ .

### C. Admitting current positive and negative evidence

The local existence increment is

$$\delta_j = \text{logit}(r_j^+) - \text{logit}(r_j^-). \quad (6)$$

The local tracker incorporates a detection score  $s$  through a fitted positive-to-negative mark ratio  $L(s) > 0$ . Belief propagation approximates the marginal association probabilities [26]. Let  $W_{jm}$  be the normalized local association mass for assigning this Bernoulli to measurement  $m$ , and let  $A_j = \sum_m W_{jm}$ . For a current sensing opportunity, use

$$g_j^+ = \sum_m W_{jm} \max\left(0, \frac{L(s_m) - 1}{L(s_m) + 1}\right), \quad (7)$$

$$g_j^- = (1 - A_j) \frac{p_{D,j}}{2 - p_{D,j}}. \quad (8)$$

Both gates are zero without a current opportunity. The first admits positive increments in proportion to association-weighted mark discrimination; the second weights the missed-detection branch by its bounded likelihood contrast. They satisfy  $g_j^+ \leq A_j$ ,  $g_j^- \leq 1 - A_j$ , and hence  $g_j^+ + g_j^- \leq 1$ .

If at least two eligible sources represent the label and none of the eligible existence inputs is a censor, define

$$\kappa_j = (1 - \beta_j) \begin{cases} g_j^+, & \delta_j \geq 0, \\ g_j^-, & \delta_j < 0. \end{cases} \quad (9)$$

Otherwise set every  $\kappa_j = 0$ . This restriction avoids constructing an unavailable current ratio for an absent label. The no-score ablation replaces the contrast in (7) by one, retaining the same score-aware local tracker and setting  $g_j^+ = A_j$ .

### D. Joint existence and spatial normalization

For each admitted source, multiply the same fractional Bernoulli ratio  $(f_j^+/f_j^-)^{\kappa_j}$  into both hypotheses in (5). The resulting spatial product is

$$h(x) = \prod_j p_j^+(x)^{\alpha_j} \prod_j \left[ \frac{p_j^+(x)}{p_j^-(x)} \right]^{\kappa_j}, \quad I = \int h(x) dx. \quad (10)$$

**Bernoulli normalization identity.** For  $0 < r_j^-, r_j^+ < 1$  and  $0 < I < \infty$ , normalizing these empty and singleton masses yields

$$p^*(x) = h(x)/I, \quad (11)$$

$$\text{logit}(r^*) = \sum_j \beta_j z_j + \sum_j \kappa_j \delta_j + \log I. \quad (12)$$

*Proof.* The empty mass is  $H_0 = A_0 \prod_j [(1 - r_j^+)/(1 - r_j^-)]^{\kappa_j}$ . The integrated singleton mass is  $H_1 = I \prod_j (r_j^+)^{\beta_j} \prod_j (r_j^+/r_j^-)^{\kappa_j}$ . Taking  $\log(H_1/H_0)$  gives (12); normalizing the singleton shape gives (11).  $\square$

The same admitted product therefore determines both spatial density and existence. GCE recomputes its spatial pool and integral, while scalar correction retains their base values in (12).

### E. Gaussian correction and curvature guard

For  $p = \mathcal{N}(m, P)$ , write

$$\log p(x) = -\frac{1}{2} x^\top J x + v^\top x + c, \quad J = P^{-1}, \quad v = Jm, \quad (13)$$

where  $c = -[d \log(2\pi) + \log |P| + m^\top Jm]/2$  and  $d = 4$ . A source transmits the differences  $(\Delta J_j, \Delta v_j, \Delta c_j)$  between its update and prediction. The four-dimensional Gaussian ratio requires fifteen numbers: ten for the symmetric precision, four for the information vector, and one constant.

Moment projection can widen a covariance after ambiguous association, so this Gaussian ratio need not be log-concave. GCE admits a source correction only when its precision increment is positive semidefinite, up to numerical tolerance:

$$\bar{\kappa}_j = \kappa_j \mathbf{1}[\Delta J_j \succeq 0]. \quad (14)$$

The guard acts on the entire ratio, including the scalar

increment. The aggregate natural parameters are

$$J = \sum_j \alpha_j J_j^+ + \sum_j \bar{\kappa}_j \Delta J_j, \quad (15)$$

$$v = \sum_j \alpha_j v_j^+ + \sum_j \bar{\kappa}_j \Delta v_j, \quad (16)$$

$$c = \sum_j \alpha_j c_j^+ + \sum_j \bar{\kappa}_j \Delta c_j. \quad (17)$$

For positive definite  $J$ , the normalized quantities are

$$P^* = J^{-1}, \quad m^* = J^{-1}v, \quad (18)$$

$$\log I = c + \frac{1}{2}[d \log(2\pi) - \log |J| + v^\top J^{-1}v]. \quad (19)$$

Use  $\bar{\kappa}$  in (12) as well. A nonfinite or ill-conditioned aggregate, or a failed Cholesky factorization, restores the base. In exact arithmetic, positive semidefinite admitted precision gains and a positive definite base ensure a positive definite aggregate. Positive semidefiniteness does not establish that an increment is statistically reliable: correlated measurement errors can pass the same test. The no-curvature ablation removes the source test while retaining the aggregate integrability check. Current ratios and gates are recomputed after each local update. For state dimension  $d$  and  $M$  eligible sources, the dense Gaussian arithmetic costs  $O(Md^3)$  and adds  $O(d^2)$  numbers per track.

**Current-likelihood interpretation.** Suppose all sources share a prior  $f^-$ , the supports and weights coincide ( $\alpha = \beta$ ), and each exact local update satisfies  $f_j^+(X) \propto f^-(X)\ell_j(X)$  for a Bernoulli set  $X$ . Substitution into the joint product gives

$$f^*(X) \propto f^-(X) \prod_j \ell_j(X)^{\omega_j}, \quad \omega_j = \beta_j + \bar{\kappa}_j \in [\beta_j, 1]. \quad (20)$$

The base retains one prior because  $\sum_j \beta_j = 1$ ; each ratio cancels its own prior. Admission therefore restores a bounded fraction of current likelihood without increasing the exponent of the inherited prior. Zero admission gives the original pool. In the formal unit-admission limit, independent observations recover the shared-prior Bayesian combination [4]. With unequal inherited beliefs, GCE uses the directly available local ratios rather than assuming a common prior can be reconstructed. This distinction motivates separating the history switch from the current-evidence gates.

#### F. Exact zero-vector transmission

The full Gaussian implementation extends a 232-byte scalar Bernoulli summary by fifteen 64-bit numbers, giving 352 bytes per track plus a 32-byte packet header. Many local updates leave all fifteen differences exactly zero. A two-byte tag distinguishes a full vector from an omitted zero vector and preserves the sign bit of each omitted zero. All nonzero vectors retain their original 120 bytes. For  $n$  tracks and  $k$  nonzero vectors, the actual packet length is

$$B = 32 + 234n + 120k \text{ bytes.} \quad (21)$$

This codec has no threshold or quantization parameter. Decoded numerical values, and hence the fusion rule, are unchanged.

## IV. EXPERIMENTS

### A. Data and evaluation scope

We use DMSTrack's released per-vehicle detections [11] on V2V4Real [10]. Nine test-backed validation sequences form the development subset (1,993 paired frames); 25 train sequences form the larger comparison (5,601 frames). Both groups and earlier train probes informed fusion design, and the detector was trained on train. These are previously used data.

An expanded evaluation covers 43 deduplicated V2V4Real segments: 9,699 paired frames from 17 recordings. It includes the earlier 9/25 cohorts and three subsequently acquired validation segments. Only one of those three, with 409 frames, comes from a previously absent recording. The checkpoint configuration assigns train/test to training/validation; its complete history is unverified.

We also evaluate all five paired-vehicle segments in the official 64-beam V2X-Real validation release [27]: 619 paired frames from three collection dates. A single-vehicle segment is excluded by input structure before scoring. Detector, calibration, tracker and communication settings remain fixed. V2X scores were first inspected after development selection.

### B. Shared local tracking and score likelihood

We track planar car centers within 40 m of either sensor and the annotated ego-frame rectangle  $[-70.4, 70.4] \times [-40, 40]$  m, excluding a 3 m disk around each vehicle. Truth and estimates use the same domain; each local sensor uses its own 40 m support. Detections retain the released confidence cutoff of 0.20.

All methods share the score-aware LBP-LMB local update. The primary constant-velocity model uses current ego coordinates, a 0.1 s interval, survival probability 0.99, acceleration intensity  $25 \text{ m}^2/\text{s}^3$ , position covariance  $I_2 \text{ m}^2$ , and in-domain  $p_D = 0.9$ . Poisson clutter has mean three per sensor scan and density  $3/(\pi 40^2)$ . Each previous-scan detection creates an independent local birth with existence 0.01, zero velocity, and covariance  $\text{diag}(16, 16, 225, 225)$ . We prune  $r \leq 0.001$ , moment-project retained mixtures to one Gaussian, and extract MAP cardinality. Geometric methods share history qualification and absent-label censoring, with in-support censor existence 0.001.

Following post-hoc probability calibration [28], we fit  $u(s) = \sigma(a \logit s + b)$ ,  $a \geq 0$ , by sequence-balanced binary log loss with regularization  $0.001a^2$ . Offline per-sensor one-to-one assignment within 12 m supplies labels. Development sequences use leave-one-sequence-out fits; the 25-sequence corpus and all new data use the full-nine fit,  $a = 2.8858$ ,  $b = 2.1564$ . The score enters each local association likelihood once through

$$L(s) = \frac{u(s)}{1 - u(s)} \frac{1 - \pi}{\pi}, \quad (22)$$

where  $\pi$  is the sequence-balanced fitting prevalence.

TABLE I

COMPARISON ON 25 PREVIOUSLY USED SEQUENCES WITH A SHARED LOCAL TRACKER. OSPA IS IN M; MISSED/FALSE GOSPA COSTS ARE IN M<sup>2</sup>.  
LOWEST VALUES ARE BOLD.

| Method          | Reliable links |               |               | Intermittent links |               |               |
|-----------------|----------------|---------------|---------------|--------------------|---------------|---------------|
|                 | OSPA           | Missed        | False         | OSPA               | Missed        | False         |
| Local           | 5.914          | 190.030       | 45.556        | 5.914              | 190.030       | 45.556        |
| MIL-AM          | 4.014          | 74.566        | 56.973        | 4.163              | 82.428        | 58.363        |
| TC-5            | 4.253          | <b>68.620</b> | 75.289        | 4.693              | 101.053       | 64.733        |
| TC-10           | 4.414          | 88.338        | 65.001        | 4.829              | 116.571       | 57.691        |
| No-age KLA      | 3.702          | 85.564        | 40.715        | 3.988              | 92.977        | 44.776        |
| Recency         | 3.672          | 78.330        | 43.986        | 3.952              | 87.113        | 47.603        |
| Conservative    | 3.672          | 86.046        | 37.436        | 3.956              | 93.865        | 42.483        |
| Capped-score    | 3.666          | 83.065        | 39.074        | 3.913              | 90.189        | 43.663        |
| Capped-cal.     | 3.650          | 80.776        | 40.418        | 3.897              | 88.391        | 44.318        |
| Scalar          | 3.526          | 78.625        | <b>34.536</b> | 3.832              | 86.342        | <b>40.523</b> |
| Guarded Scalar  | 3.508          | 75.839        | 35.438        | 3.793              | 83.328        | 41.019        |
| Fixed Ratio (0) | 3.672          | 86.046        | 37.436        | 3.956              | 93.865        | 42.483        |
| <b>GCE</b>      | <b>3.456</b>   | 71.095        | 36.303        | <b>3.761</b>       | <b>80.344</b> | 41.971        |

TABLE II

RECURSIVE CONTROLS ON THE SAME 25 SEQUENCES. SCALAR, GUARDED SCALAR, W/O CURVATURE, AND GCE FORM THE SCALAR/JOINT × GUARD-OFF/ON COMPARISON. OTHER ROWS REMOVE ONE ADMISSION COMPONENT; UNITS FOLLOW TABLE I.

| Method                | Reliable links |               |               | Intermittent links |               |               |
|-----------------------|----------------|---------------|---------------|--------------------|---------------|---------------|
|                       | OSPA           | Missed        | False         | OSPA               | Missed        | False         |
| Scalar reference      | 3.526          | 78.625        | <b>34.536</b> | 3.832              | 86.342        | <b>40.523</b> |
| Guarded Scalar        | 3.508          | 75.839        | 35.438        | 3.793              | 83.328        | 41.019        |
| w/o curvature guard   | 3.491          | 74.845        | 35.376        | 3.805              | 84.047        | 41.615        |
| w/o history switch    | 3.467          | 70.282        | 38.148        | 3.776              | 80.266        | 43.105        |
| w/o score constraint  | 3.713          | <b>55.993</b> | 59.014        | 3.922              | <b>69.042</b> | 57.784        |
| <b>GCE (complete)</b> | <b>3.456</b>   | 71.095        | 36.303        | <b>3.761</b>       | 80.344        | 41.971        |

### C. Communication and baselines

Reliable links deliver both directed packets each scan. Intermittent links combine independent directed loss of 0.1 with a bidirectional outage over the middle 20% of scans. Methods share the delivery trace and one packet opportunity per direction per scan. Receivers pool local and delivered neighbor updates with weights  $[0.5, 0.5]$ ; otherwise they keep the local update. Accounting uses 16,384-byte fragments and a 256-byte control charge per scan. Delivery is independent of packet length. GCE uses  $\rho = 0.25$  and  $\lambda = 5$  s.

*No-age KLA*, the primary reference, geometrically pools qualified posteriors using spatial and existence weights  $\alpha, b$ . *Recency* replaces  $b$  with  $q$  in existence, and *Conservative* uses  $\beta$  without a current correction. *Capped-score/Capped-cal.* bound recency existence by  $\min(r_{\text{rec}}, \max(r_{\text{noage}}, U))$ , where  $U$  is the largest association-weighted raw/calibrated score among currently observed, upweighted sources with existence at least 0.5.

*Scalar* retains the original spatial pool  $p_0$  and integral  $I_0$  while applying GCE's scalar gates without a curvature test. *Guarded Scalar* (GS) additionally uses the identical source curvature rejection and aggregate fallback as GCE on its own inputs, but returns  $p_0$  and logit  $r = \sum_j \beta_j z_j + \sum_j \bar{\kappa}_j \delta_j +$

$\log I_0$ . GS runs its own full recursion. *Fixed Ratio* retains GCE's joint update, guard conditions, and history weights, replacing both sign-dependent supports by  $\eta$ . The lowest two-condition mean OSPA on the nine development sequences selects  $\eta = 0$  from  $\{0, 0.05, 0.1, 0.125, 0.25, 0.5, 1\}$ , then holds it fixed. Zero strength retains the history-weighted base pool. GS and Fixed Ratio require full Gaussian ratio packets.

*MIL-AM* adapts the public manuscript's augmented assignment and common/exclusive subspace fusion [29]. Symmetric KL cost with unmatched cost 50 per side aligns labels; geometric methods share this alignment. Common labels use arithmetic existence pooling and existence-weighted spatial mixtures; exclusive labels retain their source density. *TC-5/TC-10* use the author's kinematic matching and two-stage multi-neighbor functions [23], with five-/ten-scan histories, order one, and cutoff 12 m. They retain separate local filters. *Local* omits communication.

### D. Metrics and recursive controls

We use  $p = 2, c = 12$  m for position OSPA [30] and missed/false squared GOSPA costs [31] ( $\alpha = 2$ ). Common-target RMSE pools robot-time-truth matches shared by

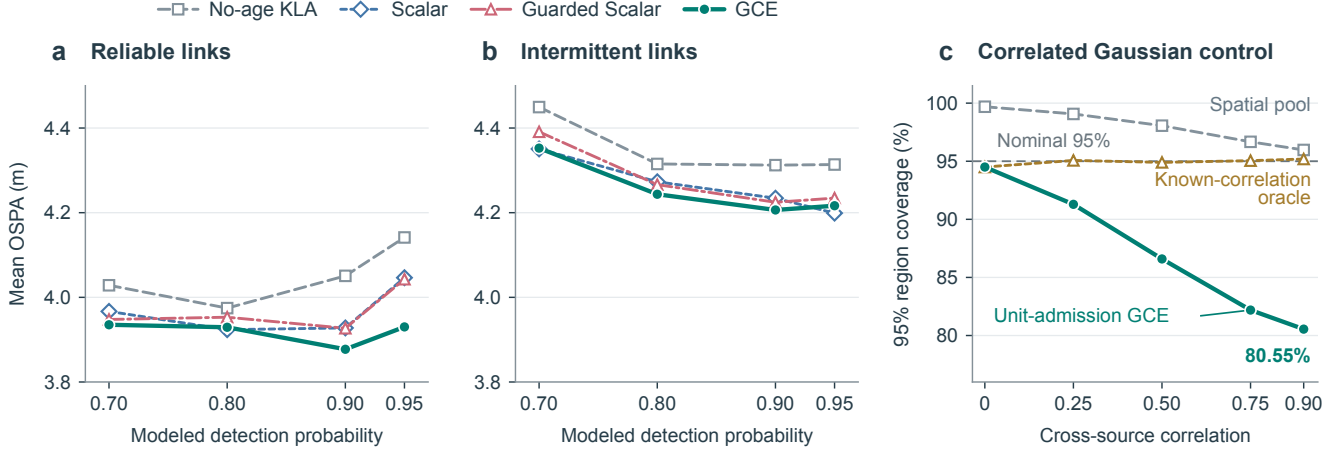


Fig. 3. Model sensitivity. (a,b) Mean OSPA over the same nine development sequences at every modeled detection probability, using all four recursive methods and identical measurements. The two panels share their vertical scale. (c) Coverage of nominal 95% regions in a conditional-existing Gaussian control with 10,000 samples at each correlation. All source curvature tests pass. The horizontal gray dashed line marks nominal coverage; unit-admission GCE increasingly undercovers as correlation grows.

both methods. Sequences receive equal weight. Descriptive 95% intervals use 10,000 paired whole-sequence percentile resamples of the old 9/25 cohorts, without multiplicity adjustment. For the expanded V2V corpus, we first average paired differences within each recording, then resample the 17 recording means. V2X results are descriptive because only three collection dates are represented.

Scalar, GS, joint correction without the guard, and GCE form a complete recursive correction–curvature  $2 \times 2$  comparison. We report GCE–GS and the interaction (GCE – GS) – (joint without guard – Scalar). Further ablations remove the history switch or positive score constraint, retaining the shared local score model. A separate fixed-input diagnostic substitutes base/corrected means and integrals at identical GCE-visited inputs; alternate outputs never feed back into its recursion.

#### E. Motion, detection probability, and correlation

Raw absolute poses match every old sequence and reproduce released relative sensor transforms. After CV prediction and birth creation, known planar yaw/translation maps positions, velocities, and full covariances to the current ego frame, equivalently implementing planar world-frame CV. No-age, Scalar, GS, and GCE run on all 34 old and three new segments. Separately, modeled  $p_D \in \{0.7, 0.8, 0.9, 0.95\}$  varies on the nine development sequences. Both checks retain the same measurements, domains, calibration, and fusion settings, without primary reselection.

To distinguish integrability from statistical reliability, we also simulate a two-dimensional, conditional-existing Gaussian state with prior covariance  $25I_2$  and two measurements of noise covariance  $I_2$ . Their cross-covariance is  $\varrho I_2$ , for  $\varrho \in \{0, 0.25, 0.5, 0.75, 0.9\}$ , with 10,000 samples per setting. We compare equal posterior pooling, unit-admission GCE assuming independent errors, and the oracle with known joint

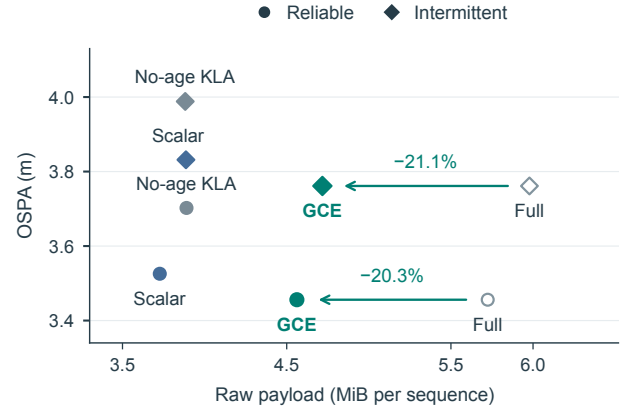


Fig. 4. Communication and tracking accuracy on the 25-sequence corpus. Circles/diamonds denote reliable/intermittent links; open markers are full GCE packets. Arrows show exact zero omission reducing raw payload at identical OSPA. Encoded GCE still costs more than No-age KLA. Lower and leftward are better.

covariance, reporting RMSE, normalized squared error, and nominal 95% region coverage. This control does not estimate the real tracker’s covariance coverage.

## V. RESULTS

### A. Tracking accuracy and adaptive admission

On the 25-segment comparison corpus, GCE has the lowest mean OSPA (Table I), improving on No-age KLA by 6.7%/5.7% and on Recency by 5.9%/4.8% (reliable/intermittent). Against No-age, missed-target cost falls by 16.9%/13.6% and false-target cost by 10.8%/6.3%. Common-target RMSE decreases from 0.964 to 0.907 m and from 1.085 to 1.050 m. Shared local tracking and delivery traces make these controlled comparisons of the fusion layers.

TABLE III  
MEAN COMMUNICATION PER SEQUENCE (MiB). FRAG. INCLUDES  
PACKET FRAGMENTATION AND CONTROL TRAFFIC.

| Method               | Reliable links |       | Intermittent links |       |
|----------------------|----------------|-------|--------------------|-------|
|                      | Raw            | Frag. | Raw                | Frag. |
| No-age KLA           | 3.89           | 8.35  | 3.88               | 8.29  |
| Scalar               | 3.73           | 8.19  | 3.89               | 8.32  |
| GCE (full)           | 5.72           | 9.79  | 5.98               | 9.95  |
| <b>GCE (encoded)</b> | 4.56           | 8.69  | 4.72               | 8.88  |

TABLE IV  
EXPANDED-CORPUS OSPA (M), WITH EQUAL SEGMENT WEIGHTS.  
V2V4REAL: 43 DEDUPLICATED SEGMENTS, 9,699 PAIRED FRAMES, 17  
RECORDINGS. V2X-REAL: FIVE SEGMENTS, 619 PAIRED FRAMES,  
THREE COLLECTION DATES. R/I DENOTE RELIABLE/INTERMITTENT  
LINKS.

| Method          | V2V4Real     |              | V2X-Real     |              |
|-----------------|--------------|--------------|--------------|--------------|
|                 | R            | I            | R            | I            |
| No-age KLA      | 3.895        | 4.144        | <b>8.359</b> | <b>8.292</b> |
| Recency         | 3.914        | 4.145        | 8.418        | 8.333        |
| Scalar          | 3.730        | 4.014        | 8.503        | 8.445        |
| Guarded Scalar  | 3.722        | 3.983        | 8.441        | 8.359        |
| w/o curvature   | 3.695        | 3.985        | 8.376        | 8.440        |
| Fixed Ratio (0) | 3.854        | 4.109        | 8.464        | 8.340        |
| <b>GCE</b>      | <b>3.671</b> | <b>3.960</b> | 8.408        | 8.338        |

GCE-Fixed Ratio is -0.216/-0.195 m. Both paired intervals lie below zero. This supports adaptive score/miss admission relative to the zero-strength control selected by the expanded seven-point grid.

### B. Joint correction and curvature control

Guarded Scalar applies the same curvature protection while retaining scalar-only correction (Table II). GCE-GS is -0.053/-0.032 m, with intervals  $[-0.10424, -0.00522]$  and  $[-0.06517, -0.00006]$ . The intermittent upper bound is within  $10^{-4}$  m of zero. On the nine development sequences, the intermittent GCE-GS interval includes zero.

The correction-curvature interaction is -0.018/-0.005 m, with intervals  $[-0.064, 0.026]$  and  $[-0.065, 0.048]$ . Both intervals include zero; this interaction contrast does not isolate a normalizer-only benefit. Removing curvature allows large position shifts and activates aggregate fallback twice. Removing positive score support trades fewer misses for more false targets; the history and score-removal total-OSPA intervals include zero.

On fixed GCE inputs, changing only the mean at corrected existence reduces immediate OSPA by about 0.003 m. The base/base-to-corrected/corrected difference is -0.014/-0.010 m, and its intervals include zero, as do those for changing the integral at corrected means. These substitutions hold admission fixed and do not alter subsequent predictions, unlike the recursive GS comparison.

### C. Expanded V2V evaluation and V2X transfer

On all 43 V2V4Real segments, GCE reduces OSPA relative to No-age by 5.8%/4.4% (Table IV). The recording-equal GCE-GS differences are -0.048/-0.017 m, with intervals  $[-0.085, -0.012]/[-0.044, 0.006]$ . The intermittent interval crosses zero; the corpus includes development inputs and related segments.

On the previously absent 409-frame V2V recording, GCE worsens OSPA to 2.789/3.055 m, compared with No-age's 2.633/3.020 m and Scalar's 2.726/2.961 m. On V2X-Real, GCE also underperforms No-age in both link conditions (Table IV). All five segments are retained, including two with nonoverlapping sensing circles.

### D. Motion and detection-probability sensitivity

With planar ego-motion compensation, GCE-GS on the 25 sequences is -0.074/-0.050 m, with paired intervals  $[-0.134, -0.023]/[-0.091, -0.016]$  below zero. Absolute error does not improve uniformly. On the 409-frame recording, GCE reaches 2.946/3.494 m, compared with No-age's 3.003/3.663 m and Scalar's 2.869/3.479 m. It beats No-age but remains worse than Scalar. Across all four modeled  $p_D$  values, GCE retains a lower mean than No-age and GS, but Scalar wins at reliable  $p_D = 0.8$  and intermittent  $p_D \in \{0.7, 0.95\}$  (Fig. 3a,b). Varying a constant  $p_D$  cannot validate visibility under occlusion.

### E. Uncertainty under correlated observations

All precision increments in the Gaussian control are positive semidefinite. At zero correlation, GCE and the known-correlation oracle agree. At  $\rho = 0.9$ , GCE's nominal 95% regions cover only 80.55% of samples, versus 95.19% for the oracle (Fig. 3c), with mean normalized squared error per dimension 1.847. Coverage with adaptive gates remains unmeasured.

### F. Communication cost

The codec preserves GCE trajectories and reduces raw payload by 20.3%/21.1%. Fragmentation and control traffic reduce this saving to 11.3%/10.8% (Table III). Relative to No-age KLA, encoded GCE requires 17.3%/21.5% more raw payload and 4.1%/7.0% more fragmented bytes. GS and Fixed Ratio use GCE's full 352-byte schema; Scalar uses 232 bytes. The loss model does not measure wireless latency, capacity, or energy.

## VI. DISCUSSION AND CONCLUSION

GCE augments a conservative posterior pool with selected current Bernoulli ratios, jointly updating spatial density and existence. On 43 V2V4Real segments, mean OSPA falls by 5.8%/4.4% relative to No-age KLA. Exact zero omission preserves the tracking outputs while saving 11.3%/10.8% of fragmented traffic relative to full GCE packets on the 25-segment corpus.

Gains beyond guarded scalar correction are smaller; V2X transfer and covariance calibration remain open. Future evaluation should cover independent recordings, larger teams and measured radio links.

## ACKNOWLEDGMENT

OpenAI Codex was used to develop the experimental and analysis code, produce the schematic figures and the result figures and tables from saved outputs, and draft and revise Sections I–VI, the abstract, captions, and mathematical exposition. The introductory scene and mechanism overview were designed with OpenAI image generation and reconstructed as editable vector graphics.

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